Statistical Data Analysis and Visualization Lab

Discrete Probability Distributions: Binomial, Poisson & Geometric

***Prompt 1** — Create the basic lab application*

Create a simple educational web application for an undergraduate AIML laboratory.

Title:

Discrete Probability Distributions

The application will eventually contain:

1. Binomial Distribution

2. Poisson Distribution

3. Geometric Distribution

For now, implement only the basic application structure and the Binomial Distribution module.

Keep the UI simple and clean. Do not add unnecessary features.

Create a navigation/header with:

- Home

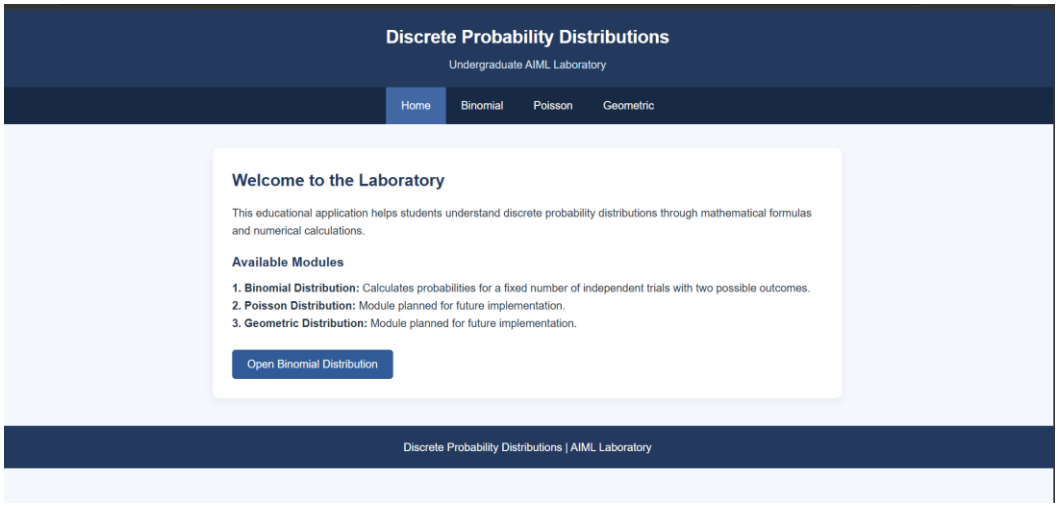
- Binomial

- Poisson

- Geometric

Only the Binomial page needs to be functional at this stage.

Output:



Discrete Probability Distributions

Undergraduate AIML Laboratory

Home | Binomial | Poisson | Geometric

Binomial Distribution

The binomial distribution calculates the probability of obtaining exactly **k successes** in **n independent trials**, when the probability of success in each trial is p.

$$P(X = k) = C(n, k)\, p^k\, (1 - p)^{n - k}$$

Where:

- n = Total number of trials
- k = Number of successes
- p = Probability of success
- 1 - p = Probability of failure
- C(n, k) = Number of combinations

Enter the Parameters

Number of trials (n)	Probability of success (p)	Number of successes (k)
5	0.5	2

Number of trials (n)	Probability of success (p)	Number of successes (k)
5	0.5	2

Calculate Probability

Calculated Probability
P(X = 2) = 0.312500
Percentage = 31.2500%

Probability Distribution Table

Successes (k)	P(X = k)	Percentage
0	0.031250	3.1250%
1	0.156250	15.6250%
2	0.312500	31.2500%
3	0.312500	31.2500%
4	0.156250	15.6250%
5	0.031250	3.1250%

Enter p as a decimal between 0 and 1. For example, 0.5 represents a 50% probability of success.

Discrete Probability Distributions

Undergraduate AIML Laboratory

Home | Binomial | Poisson | Geometric

Poisson Distribution

This module will be implemented in a future version.

Discrete Probability Distributions

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Home | Binomial | Poisson | Geometric

Geometric Distribution

This module will be implemented in a future version.

Binomial Distribution

***Prompt 2** — Add Binomial inputs*

Implement the Binomial Distribution section.

Add two input controls:

- Number of trials n

- Probability of success p

Set default values:

n = 20

p = 0.35

Validate the inputs:

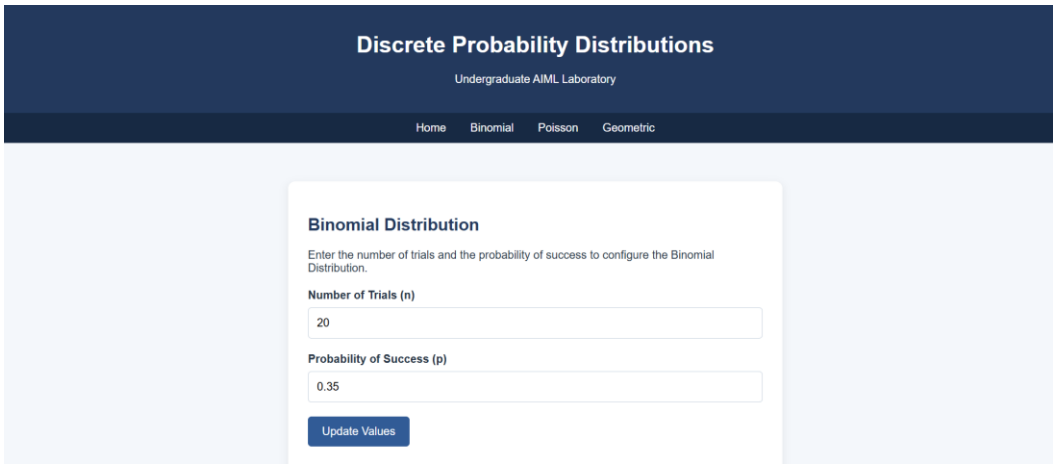
- n must be a positive integer

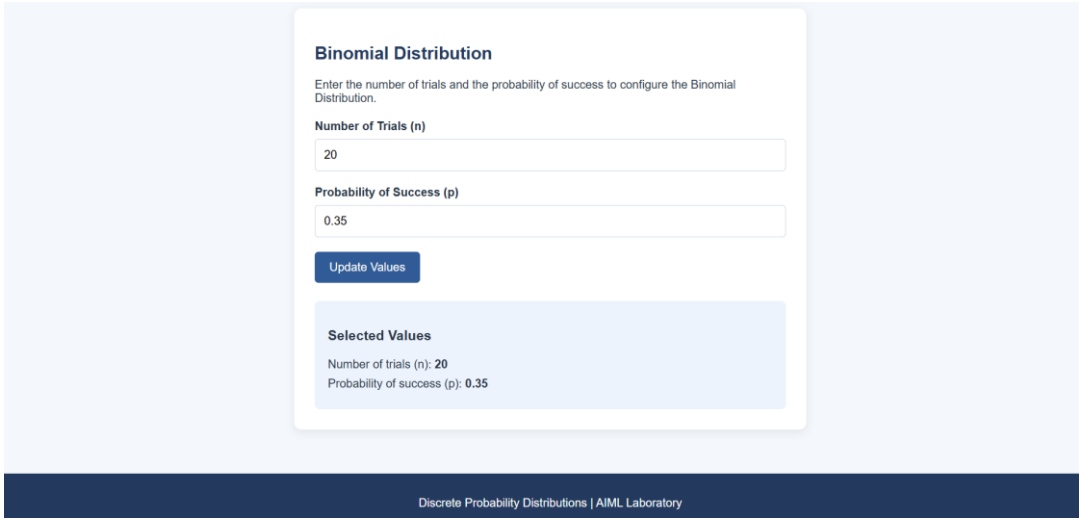
- p must be between 0 and 1

Display the selected values clearly.

Do not add simulation yet.

Output:





***Prompt 3** — Calculate Binomial statistics*

Add the Binomial probability calculations using the selected n and p.

Calculate and display:

- Mean

- Variance

- Standard deviation

- Second moment $E[X^2]$

- Skewness

Use the following formulas:

Mean:

np

Variance:

np(1-p)

Second moment:

Variance + Mean²

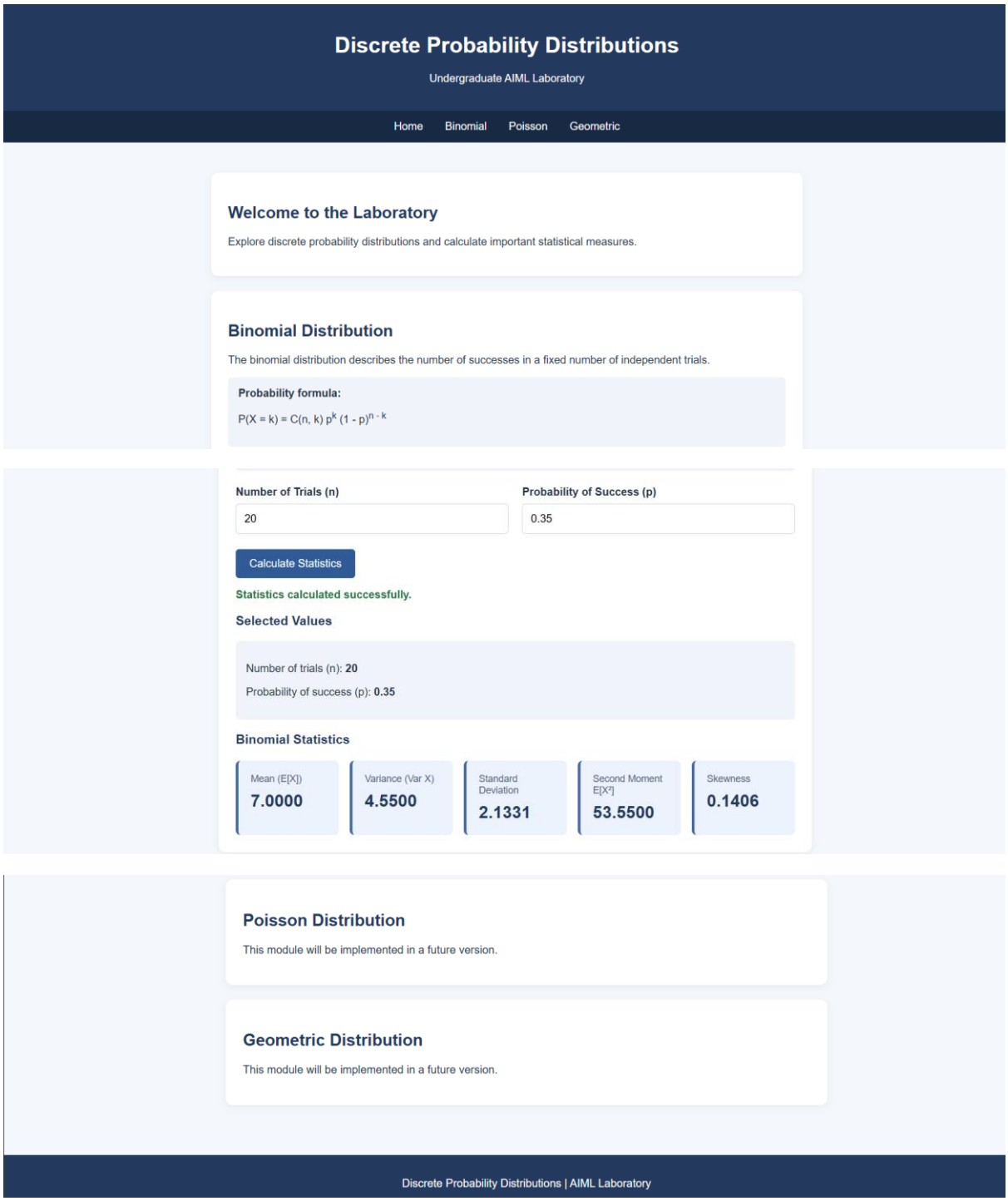
Skewness:

(1-2p) / sqrt(np(1-p))

Display the results in simple statistic cards.

For the default values n=20 and p=0.35, verify that the mean is 7 and variance is 4.55.

Output:



__Prompt 4__ — Add PMF

Add a Binomial PMF calculator.

For:

X ~ Binomial(n,p)

calculate:

P(X = k)

Allow the student to enter k.

Validate that:

0 <= k <= n

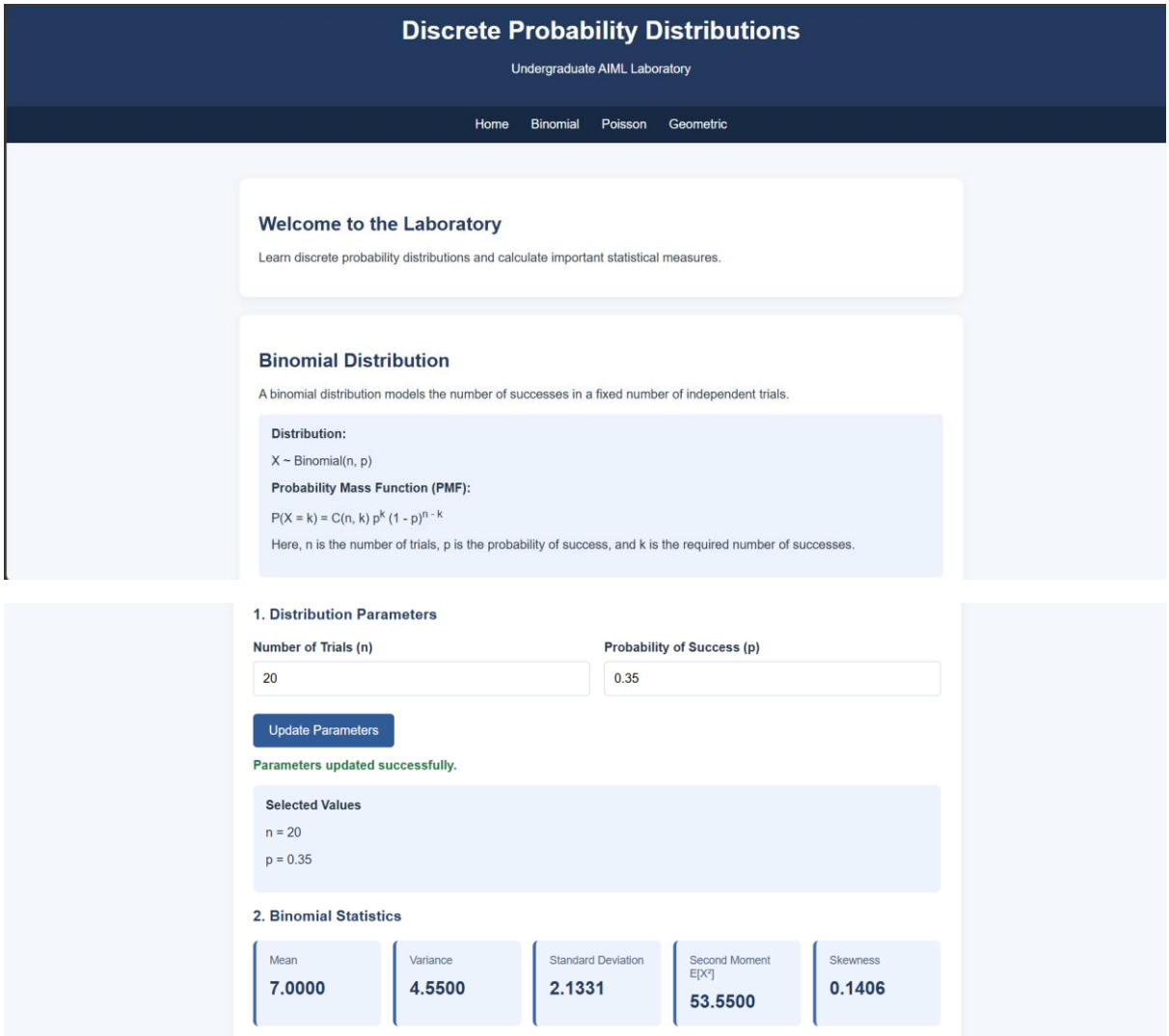
Display the probability clearly.

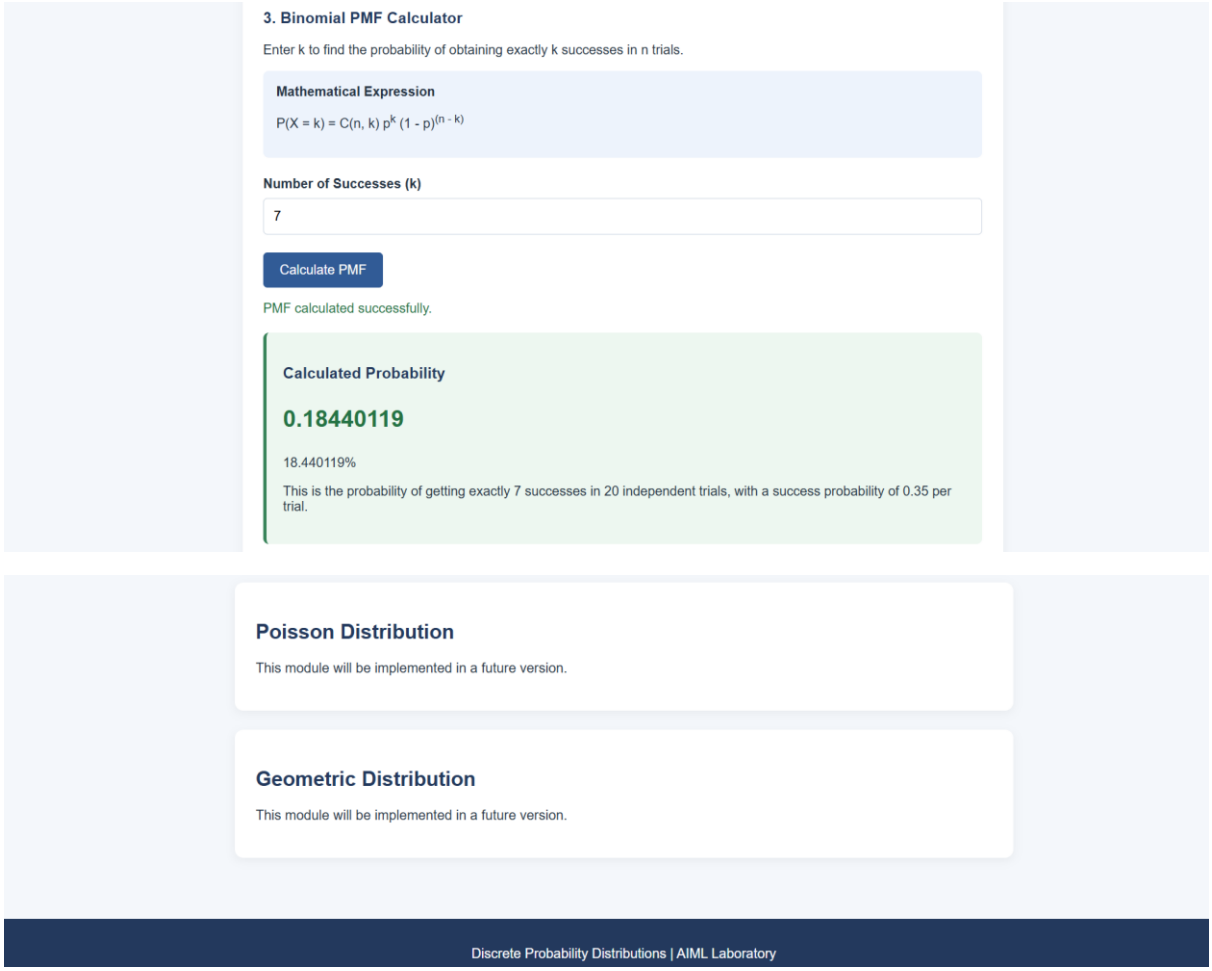
Also display the mathematical expression:

P(X=k) = C(n,k) p^k (1-p)^(n-k)

Explain in one simple sentence what the result means.

Output:





***Prompt 5** — Add CDF*

Add a Binomial CDF calculator.

Allow the student to enter k.

Calculate:

P(X <= k)

Also calculate:

P(X > k) = 1 - P(X <= k)

Display both results.

Explain:

“P(X <= k) means the probability of getting k or fewer successes.”

Output:

Discrete Probability Distributions

Undergraduate AIML Laboratory

Home Binomial Poisson Geometric

Welcome to the Laboratory

Explore discrete probability distributions and calculate important statistical measures.

Binomial Distribution

A binomial distribution models the number of successes in a fixed number of independent trials.

Distribution:

$X \sim \text{Binomial}(n, p)$

PMF:

$P(X = k) = C(n, k)\, p^k\, (1 - p)^{n - k}$

CDF:

$P(X \le k) = \sum_{i=0}^{k} P(X = i)$

$P(X > k) = 1 - P(X \le k)$

1. Distribution Parameters

Number of Trials (n)

20

Probability of Success (p)

0.35

Update Parameters

Parameters updated successfully.

Selected Values

n = 20

p = 0.35

2. Binomial Statistics

Mean	Variance	Standard Deviation	Second Moment $E[X^2]$	Skewness
7.0000	4.5500	2.1331	53.5500	0.1406

3. Binomial PMF Calculator

$P(X = k) = C(n, k)\, p^k\, (1 - p)^{n - k}$

Number of Successes (k)

7

Calculate PMF

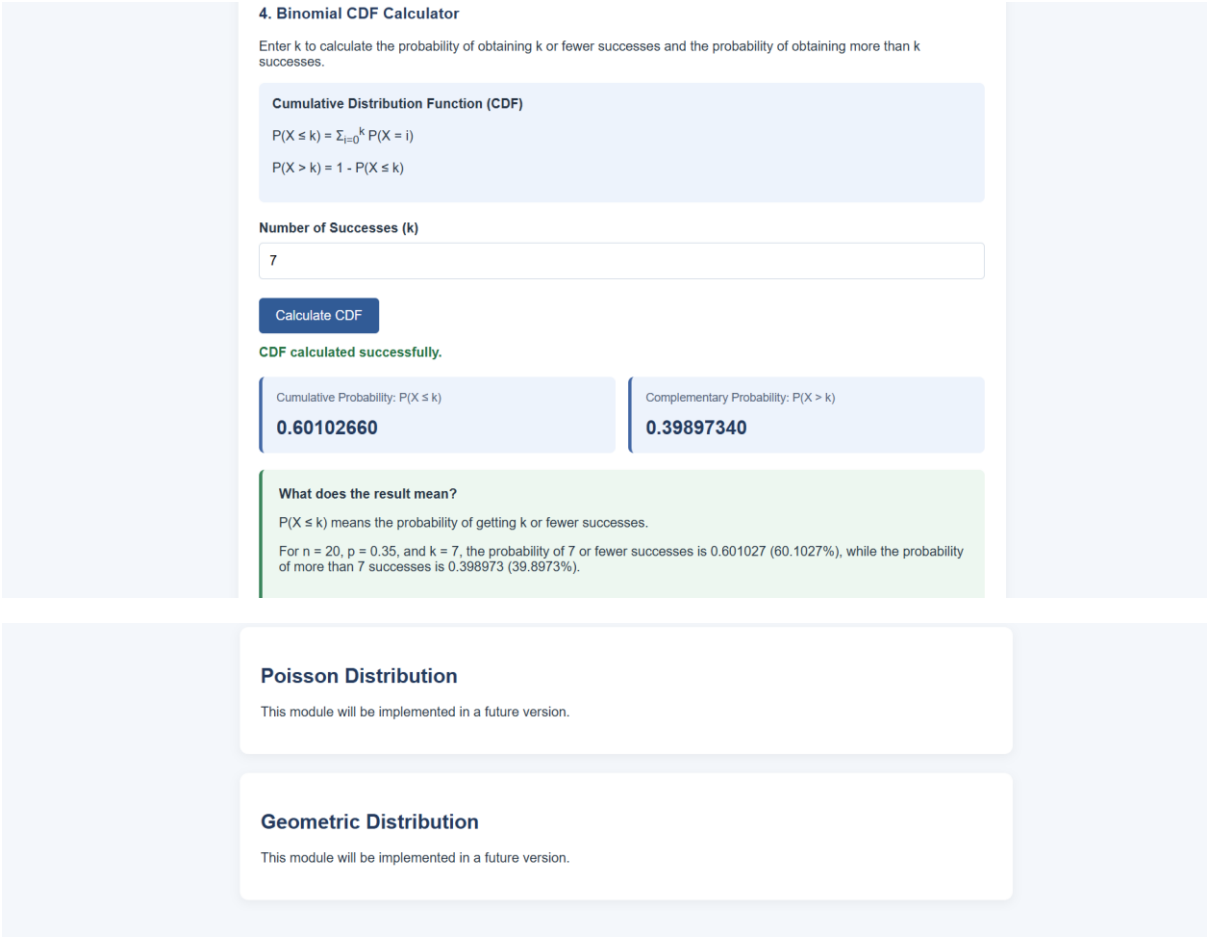
PMF calculated successfully.

PMF Result: P(X = k)

0.18440119

18.440119%

Probability of exactly 7 successes in 20 independent trials.



__Prompt 6__ — Add Binomial PMF graph

Add a bar chart showing the Binomial PMF.

X-axis:

Number of successes X

Y-axis:

Probability P(X)

Plot values for:

X = 0, 1, 2, ..., n

Use the current values of n and p.

Add a vertical dashed reference line at the mean.

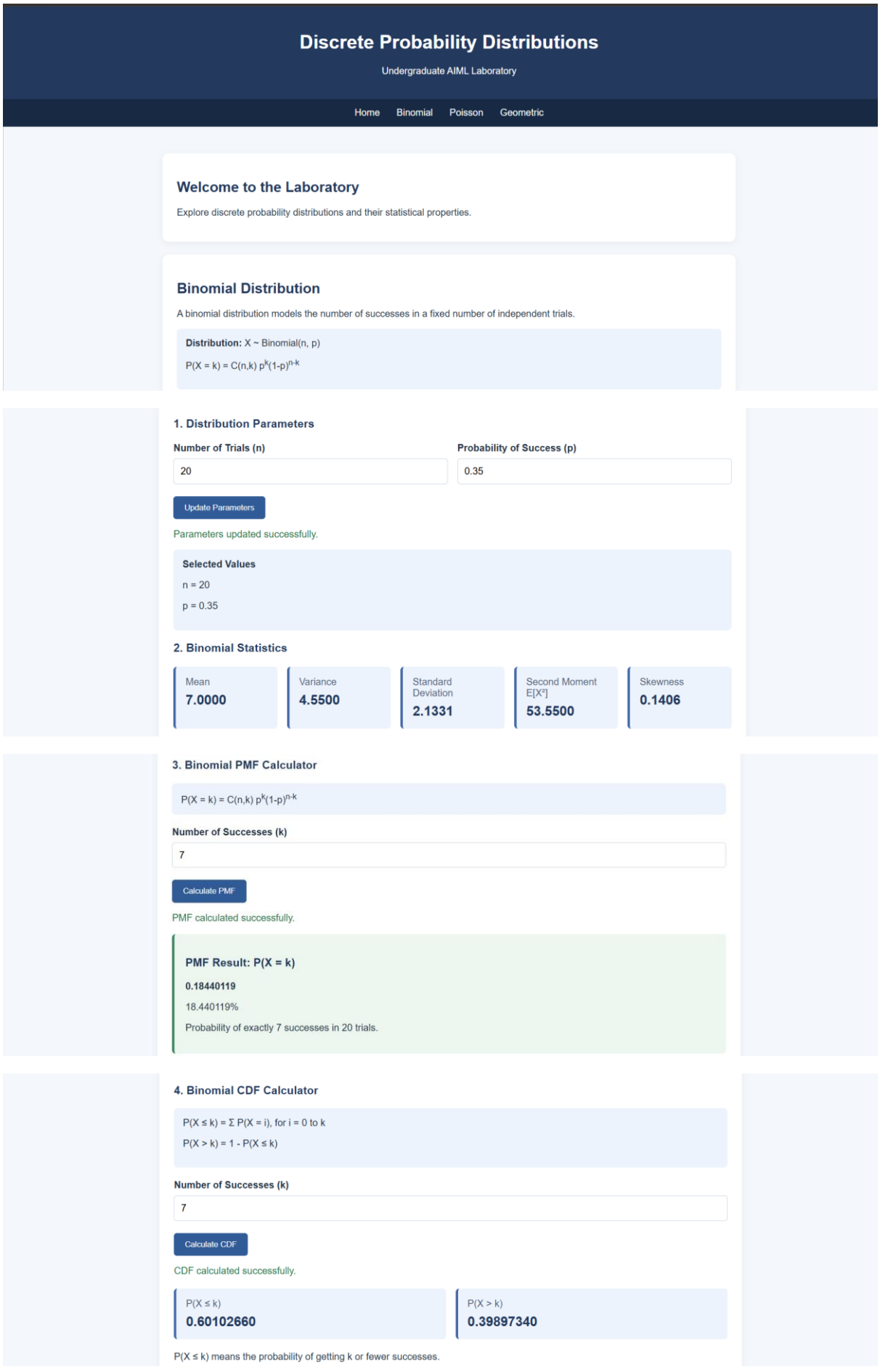
Add a tooltip showing:

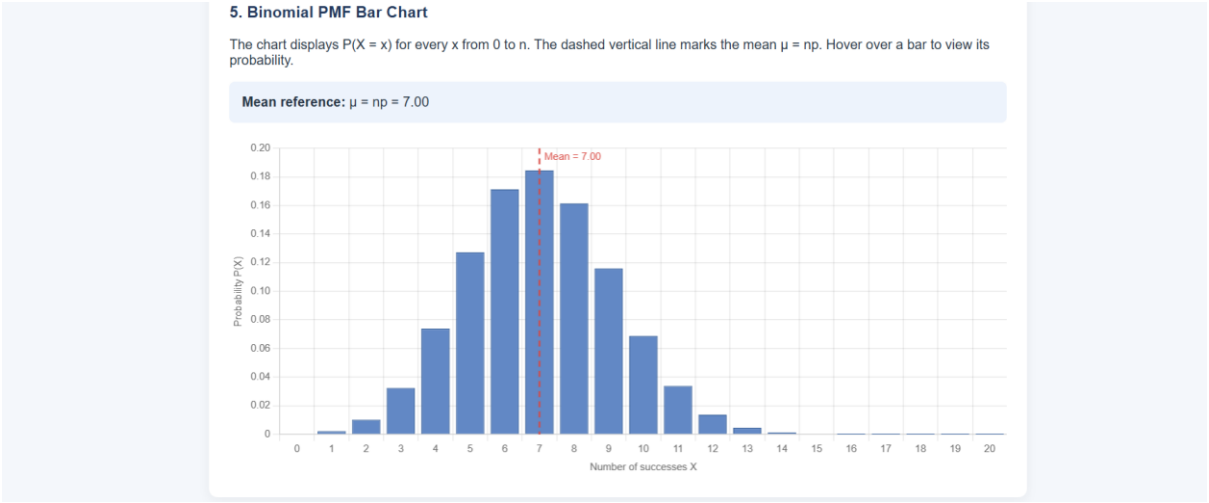
X = value

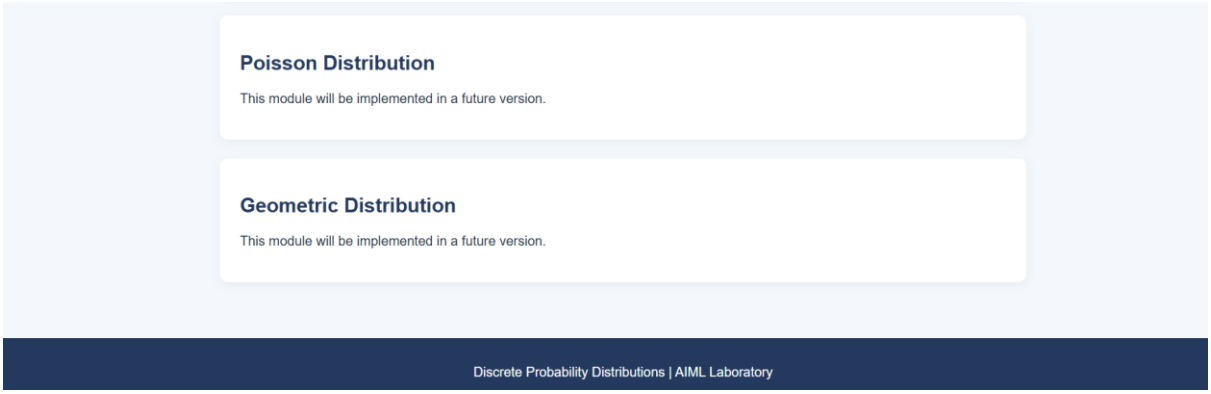
P(X=value) = probability

Update the graph automatically whenever n or p changes.

Output:


Discrete Probability Distributions
Undergraduate AIML Laboratory
Home Binomial Poisson Geometric
Welcome to the Laboratory
Explore discrete probability distributions and their statistical properties.
Binomial Distribution
A binomial distribution models the number of successes in a fixed number of independent trials.
Distribution: X ~ Binomial(n, p)
P(X = k) = C(n,k) p^k(1-p)^n-k
1. Distribution Parameters
Number of Trials (n)
20
Probability of Success (p)
0.35
Update Parameters
Parameters updated successfully.
Selected Values
n = 20
p = 0.35
2. Binomial Statistics
Mean
7.0000
Variance
4.5500
Standard Deviation
2.1331
Second Moment E[X²]
53.5500
Skewness
0.1406
3. Binomial PMF Calculator
P(X = k) = C(n,k) p^k(1-p)^n-k
Number of Successes (k)
7
Calculate PMF
PMF calculated successfully.
PMF Result: P(X = k)
0.18440119
18.440119%
Probability of exactly 7 successes in 20 trials.
4. Binomial CDF Calculator
P(X ≤ k) = Σ P(X = i), for i = 0 to k
P(X > k) = 1 - P(X ≤ k)
Number of Successes (k)
7
Calculate CDF
CDF calculated successfully.
P(X ≤ k)
0.60102660
P(X > k)
0.39897340
P(X ≤ k) means the probability of getting k or fewer successes.



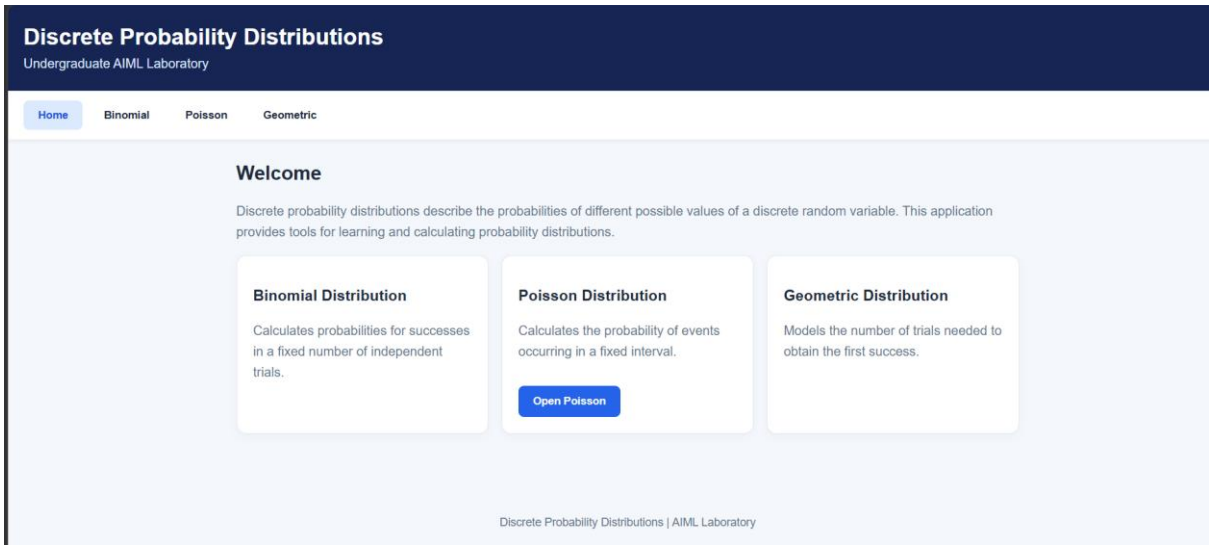


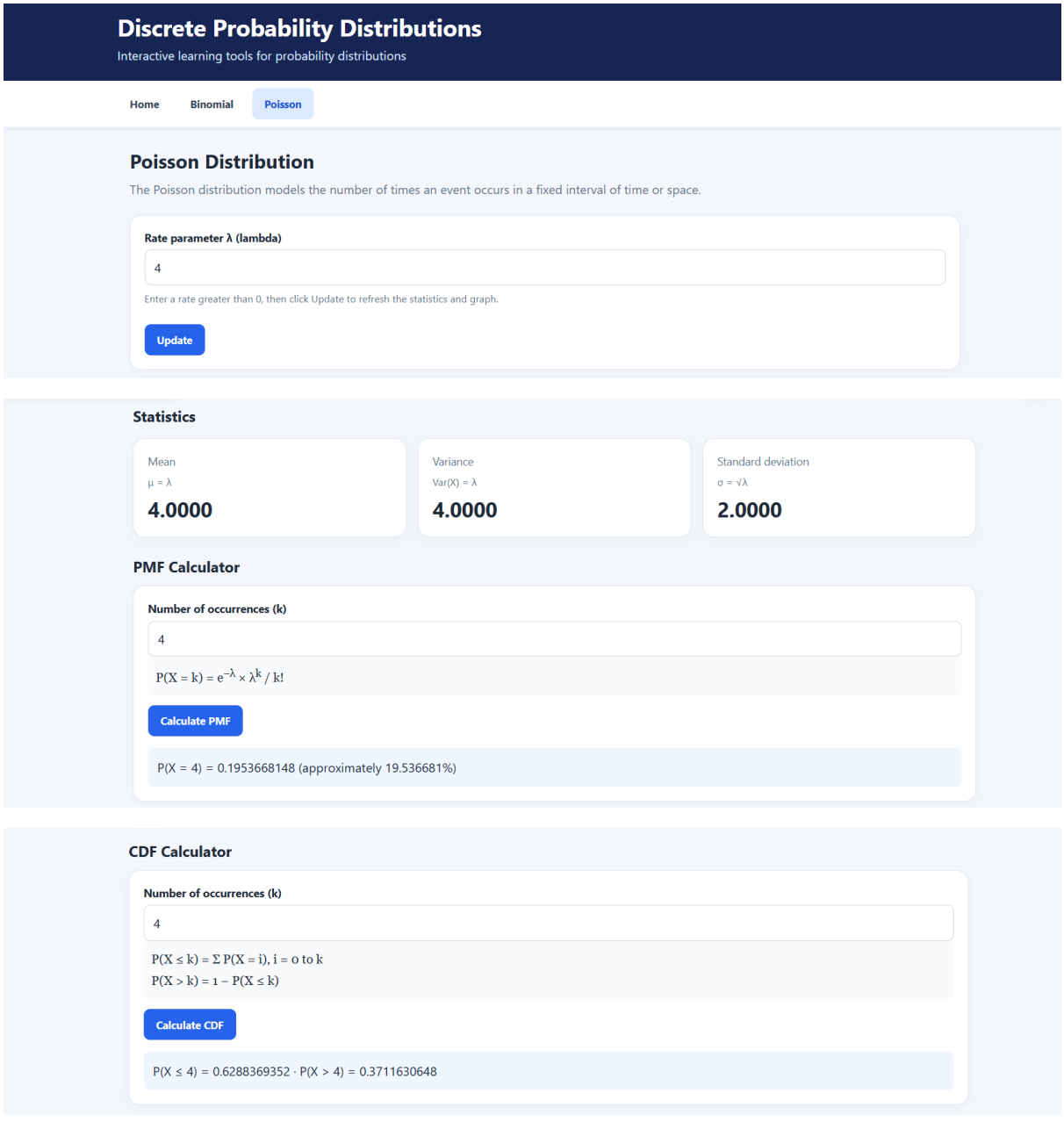
Prompt 7:

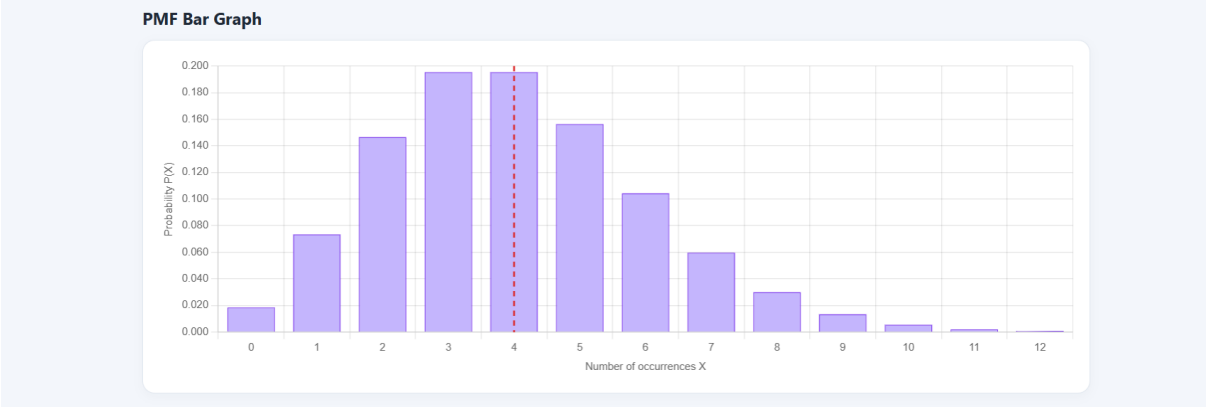
Add a Poisson Distribution module to my existing "Discrete Probability Distributions" HTML application.

- Keep the existing Binomial module fully functional.
- Keep the same navigation, design, cards, buttons, and responsive layout.
- Add Poisson to the navigation bar.
- Set the default rate parameter to $\lambda = 4$.
- λ must be greater than 0. Display an error message for invalid inputs.
- Add Input Parameters, Statistics, PMF Calculation, CDF Calculation, and a PMF bar chart.
- Calculate Mean = λ, Variance = λ, and Standard Deviation = $\sqrt{\lambda}$.
- Use PMF: $P(X=k) = (e^{-\lambda} \times \lambda^k) / k!$, where k is a non-negative integer.
- Use CDF: $P(X \leq k) = \Sigma P(X=i)$, for i = 0 to k.
- Also calculate $P(X>k) = 1 - P(X \leq k)$.
- Set the default value of k to 4.
- Display the results in cards matching the existing Binomial module.
- Use purple bars and a vertical dashed red line at the mean in the graph.
- When λ changes and the Update button is clicked, automatically update the statistics and graph.
- Validate inputs and display appropriate error messages.
- Write the actual HTML, CSS, and JavaScript in my existing file. Do not create a separate application or change the existing Binomial module.
- Return the complete updated HTML file.

Output:


Discrete Probability Distributions
Undergraduate AIML Laboratory
Home
Binomial
Poisson
Geometric
Welcome
Discrete probability distributions describe the probabilities of different possible values of a discrete random variable. This application provides tools for learning and calculating probability distributions.
Binomial Distribution
Calculates probabilities for successes in a fixed number of independent trials.
Poisson Distribution
Calculates the probability of events occurring in a fixed interval.
Open Poisson
Geometric Distribution
Models the number of trials needed to obtain the first success.
Discrete Probability Distributions | AIML Laboratory


Discrete Probability Distributions
Interactive learning tools for probability distributions
Home
Binomial
Poisson
Poisson Distribution
The Poisson distribution models the number of times an event occurs in a fixed interval of time or space.
Rate parameter λ (lambda)
4
Enter a rate greater than 0, then click Update to refresh the statistics and graph.
Update
Statistics
Mean
μ = λ
4.0000
Variance
Var(X) = λ
4.0000
Standard deviation
σ = √λ
2.0000
PMF Calculator
Number of occurrences (k)
4
P(X = k) = e^−λ × λ^k / k!
Calculate PMF
P(X = 4) = 0.1953668148 (approximately 19.536681%)
CDF Calculator
Number of occurrences (k)
4
P(X ≤ k) = Σ P(X = i), i = 0 to k
P(X > k) = 1 − P(X ≤ k)
Calculate CDF
P(X ≤ 4) = 0.6288369352 · P(X > 4) = 0.3711630648

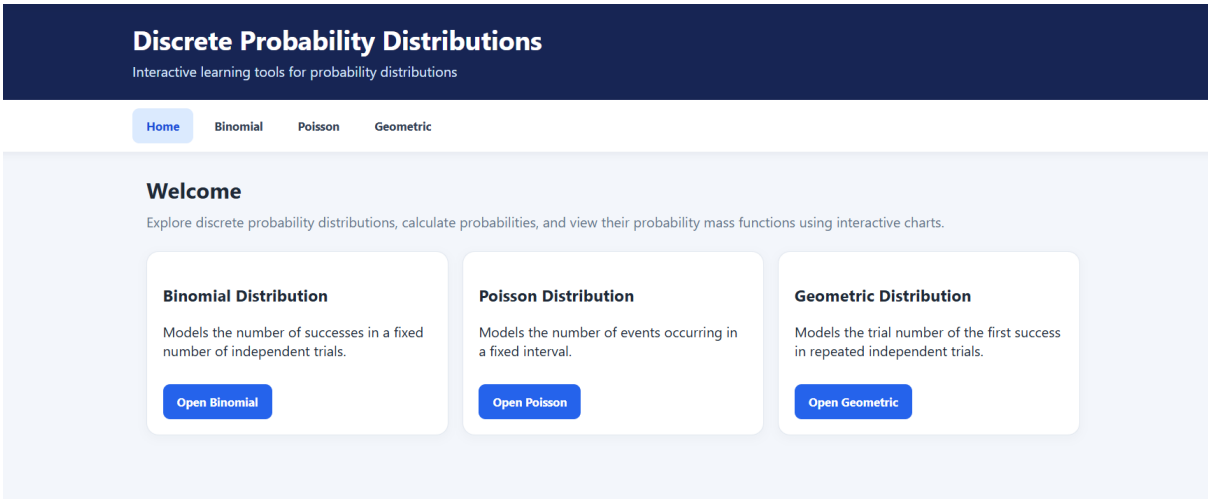


Prompt 8:

Add a Geometric Distribution module to my existing “Discrete Probability Distributions” HTML application.

- Keep the existing Binomial and Poisson modules fully functional.
- Keep the same navigation, design, cards, buttons, and responsive layout.
- Add Geometric to the navigation bar.
- Set the default probability of success to p = 0.35.
- Use the convention X = trial number of the first success, starting from 1.
- Add Input Parameters, Statistics, PMF Calculation, CDF Calculation, and a PMF bar chart.
- Calculate Mean = 1/p, Variance = $(1-p)/p^2$, and Standard Deviation = $\sqrt{(1-p)/p}$.
- Use PMF: $P(X=k) = (1-p)^{(k-1)} \times p$.
- Use CDF: $P(X \leq k) = 1-(1-p)^k$.
- Validate inputs and display error messages.
- Use orange bars and a vertical dashed red line at the mean in the graph.
- Write the actual HTML, CSS, and JavaScript in my existing file. Do not create a separate application or change the existing modules.
- Return the complete updated HTML file.

Output:



Discrete Probability Distributions

Interactive learning tools for probability distributions

Home | Binomial | Poisson | Geometric

Geometric Distribution

The geometric distribution models the trial number on which the first success occurs. Here, X starts at 1.

1. Input Parameters

Probability of success (p)

0.35

Enter a probability greater than 0 and no greater than 1, then click Update.

Update

2. Statistics

Mean	Variance	Standard deviation
$\mu = 1 / p$	$\mathrm{Var}(X) = (1 - p) / p^2$	$\sigma = \sqrt{(1 - p) / p}$
2.8571	**5.3061**	**2.3035**

3. PMF Calculation

Trial number (k), starting at 1

3

$P(X = k) = (1 - p)^{k-1} \times p$, for $k = 1, 2, 3, \ldots$

Calculate PMF

$P(X = 3) = 0.1478750000$

4. CDF Calculation

Trial number (k), starting at 1

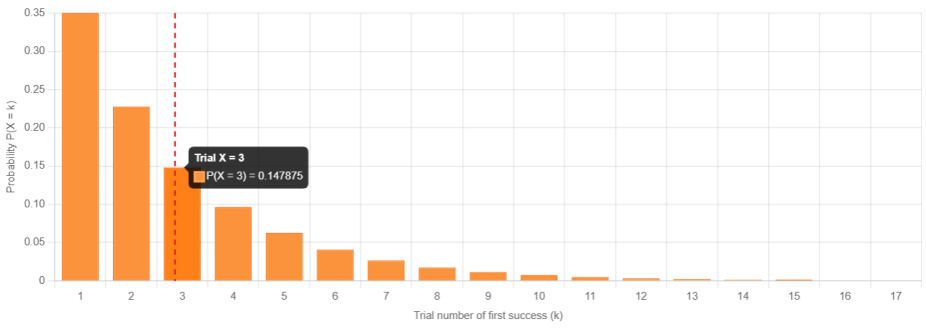
3

$P(X \le k) = 1 - (1 - p)^k$, for $k \ge 1$

Calculate CDF

$P(X \le 3) = 0.7253750000 \cdot P(X > 3) = 0.2746250000$

5. Geometric Distribution PMF Bar Chart



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